

Fluid Flow Through Fluidized Beds

Pre-Lab Plan

Unit Operations Lab 4 

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Group #2, Tuesday, PM section

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1. Experimental Objective (5 pts)

The objective of this experiment is to compare an experimentally obtained minimum fluidization velocity to the theoretical value obtained from the Ergun Equation. This will be accomplished by measuring the pressure drop across a bed of particles as a function of the compressed air flowrate. This study of fluidization behavior will use a bed of fine silica sand and MBD 15 SC ion exchange resin beads as its bed material.

2. Experimental

2.1 Apparatus (5 pts)

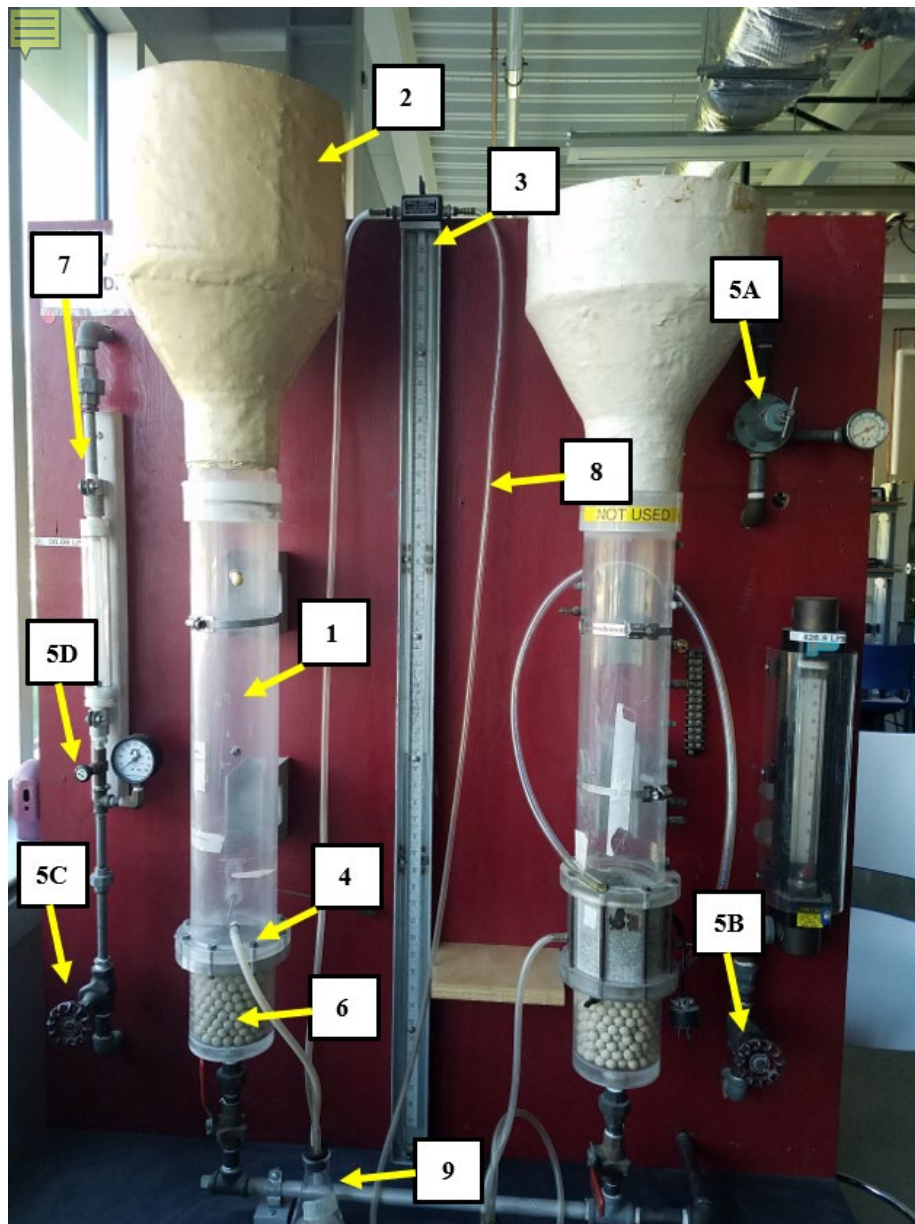


Figure 1: Picture of apparatus used during the experiment

Table 1: Description of Apparatus Used in the Experiment

Number	Instrument	Description
1	4" plexiglass tube	A clear tube which houses the bed and has compressed air flowing from the bottom to the top.
2	Loading Funnel	An enlargement at top of 4" plexiglass tube which facilitates easy loading of bed material and helps reduce entrainment losses.
3	Manometer	Device that measures the pressure differential at the bed (bottom of 4" Plexiglas tube) in inches of water.
4	Porous Metal Disk	A metal disk that supports the bed and allows air to enter from the bottom of the plexiglass tube
5A	Valve	Valve that controls the flowrate of air into the system.
5B	Valve	Valve that controls the flowrate of compressed air through the 426.8 LPM rotameter.
5C	Valve	Valve that controls the flowrate of compressed air to the 36.09 LPM rotameter.
5D	Valve	Valve that controls the flowrate of compressed air through the 36.09 LPM rotameter.
6	Fluidized Bed Vessel Base	The bottom part of the plexiglass column, contains spherical beads which creates an even air flow of the compressed air.
7	Rotameter	Measures flow rate of compressed air entering the column based on a percentage of the maximum flow rate of 36.09 std L/min.
8	Rubber Tubes	Tubes to transport compressed air through the system.
9	Flask	Collects any bedding material that is in the air before being sent to the manometer.

2.2 Materials and Supplies (5 pts)

Table 1: Description of Materials and Supplies Used in the Experiment

Materials/Supplies	Description/Purpose
Fine Silica Sand	Sand that is used as the bedding material for the experiment.
Ion Exchange Resin Beads	Ion exchange resin beads that are used for the bedding material for the experiment. MBD 15 SC Ion Exchange Resin Beads.
Sieve	Trays that have different mesh sizes that allow particles with different sizes to pass through or not pass through.
Ruler	Used to measure the 6" and 10" bed heights as well as the difference in manometer water height.
Graduated Cylinder	Used to measure the volume of the bedding material. Will be used to determine the density.
Scoop	Used to transfer the bedding material into the plexiglass tube.
Compressed Air	The fluid that will be flowing through the plexiglass tube to fluidize the bedding material.
Vacuum	Used to vacuum out the bedding material after the experiment is completed.

2.3 Experimental Plan (10 pts)

Fluidized Bed

1. Choose either the fine silica sand or the ion exchange resin as your bedding material.
2. Determine the void fraction of the material. Begin by adding 20 mL of bedding material to a graduated cylinder. Weigh and record the mass. These mass and volume values will be used to determine the density of the material to use in the Ergun equation calculations. Next, use the same volume/mass of the material and add water to the graduated cylinder until the water level reaches the 20mL mark. Record the volume of water added to reach the 20mL mark stated above.
3. Add the bedding material to the vessel to a packed height of 6" within the 4" plexiglass tube.
4. Slowly turn on the compressed air supply by opening valve 5A and adjust valve 5B to 10% of 426.8 L/min. Then, open valve 5D and use valve 5C to more accurately control the flowrate of air through the fluidized bed. Record the pressure drop and bed height at each flowrate increment by using the manometer and a ruler. The flowrate of air will be brought to specific percentages of the maximum total flowrate in increments of 5% of the flowrate discharging from valve 5D.
5. Continue to increase the flowrate of air beyond the minimization fluidization conditions and continue to record the pressure drop and bed height. Minimum fluidization is reached once the pressure drop does not increase significantly at increasing compressed air flowrates.
6. Then decrease the flowrate of the compressed air to 0 L/min in increments of 5%, making sure to record the pressure drop and bed height at each increment.
7. Repeat the procedure, making sure to do static bed height of 6" and 10" for both bedding materials.

Average Particle Diameter of Sand

1. Weight each empty sieve tray and record its weight.
2. Set up a column of sieves with the largest mesh diameter at the top of the column and the smallest mesh diameter at the bottom of the column.
3. Measure out 100 grams of sand and pour it slowly on the top sieve (largest mesh diameter). Shake the column vigorously for a few minutes to make sure complete separation has occurred.
4. Weight each sieve tray and record the final mass. Subtract the initial mass of the sieve tray from the final mass of the sieve tray to get the mass of sand on each tray.
5. Divide the mass of the sand on each tray by 100 grams to get the mass fraction of each particle size in the sand.
6. Use the following equation to determine the average particle size:

$$\langle D_p \rangle = \frac{1}{\sum (x_i / D_{p,i})}$$

Where x_i is the mass fraction of particles of size $D_{p,i}$.

Project Safety Evaluation (10 pts)

Project Title: Fluid Flow Through Fluidized Bed

Date: 10/15/2019

	Yes/No	
Potential electrical hazards?	No	
Potential mechanical hazards?	Yes	Make sure that the gas cylinder is well supported so it doesn't fall and hurt somebody.
Potential pressure hazards?	Yes	Compressed air is stored at high pressure. Buildup of pressure could burst equipment.
Potential temperature hazards?	No	
Hazardous/toxic chemicals involved?	Yes	Fine silica sand and the ion exchange resin beads can cause eye irritation.
Gloves required?	Yes	Nitrile/latex gloves to prevent the sand or resin coming in contact with skin
MSDS reviewed by all team members?	Yes	Compressed Air, Fine Silica Sand, and Ion Exchange Resin
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	23 °C	The compressed air and apparatus is at room temperature, so no temperature hazards.
Pressure (psia)	70 psig	The gas is pressurized in the plexiglass tubes and in the pipes.
Safety Controls	yes/no	If yes, Provide description
	yes	Main air supply valve, shut off if system is over pressurized.

I have read relevant background material for the Unit Operations Laboratory entitled: "Fluid Flow Through Fluidized Bed" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Jasmine Villegas 10/15/2019

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